

TABLE II
Specification Comparison of Various Instruments from Different Manufacturers

Instrument	Make	Series/Model	Specifications
Spectrum analyser	Agilent Technologies	ESA-L series E-4411B	Freq: 9 kHz to 1.5 GHz RBW: 1 kHz to 5 MHz Sensitivity: -130 dBm at 1 kHz RBW
		E-4403B	Freq: 9 kHz to 3.0 GHz RBW: 1 kHz to 5 MHz Sensitivity: -130 dBm at 1 kHz RBW
		E-4408B	Freq: 9 kHz to 26.5 GHz RBW: 1 kHz to 5 MHz All these models have RBW option of 100 Hz
		ESA-E series E-4402B E-4404B E-4405-B E-4407B	Freq: 9 kHz to 3 GHz Freq: 9 kHz to 6.7 GHz Freq: 9 kHz to 13.7 GHz Freq: 9 kHz to 26.5 GHz All these models have RBW of 1 kHz to 5 MHz and option for 10Hz and 1Hz RBW. Sensitivity is -135 dBm with 1Hz RBW
RF power meter	Boonton	RF4240 series Sensor 51071A	Freq: 10 MHz to 26.5 GHz Power: -70 to +20 dBm
		Sensor 51079A	Freq: 500 kHz to 18.0 GHz Power: -50 to +40 dBm
		Sensor 51075A	Freq: 500 kHz to 18 GHz Power: -70 to +20 dBm
		Sensor 51077A	Freq: 500 kHz to 18 GHz Power: -60 to +30 dBm The instruments have 5-digit resolution (0.001 dBm) and an accuracy of ± 0.105 dB
	Agilent Technologies	N432A power meter with 8478B power sensor	Freq: 100 kHz to 18 GHz Power: -30 dBm to +10 dBm It has a reading accuracy of 0.1 per cent and resolution of 0.001 dB
		V3500A handheld power meter with built-in power sensor	Freq: 10 MHz to 6 GHz Power: -60 to +20 dBm It has 4-digit resolution (0.01 dB) and accuracy of ± 1.97 dB (worst case)

Apart from the above specifications, some specifications for quick checking of the instruments should be specified. For example, in power meters and oscilloscopes, CAL out is available for checking the instrument. In signal generators, there may be requirement for amplitude modulation (AM), frequency modulation (FM), etc. For voltage standing-wave ratio (VSWR) measurements in transmission lines using slotted line, squarewave modulation is needed. These specifications are required in synthesised frequency generators.

Thus the various specifications should be carefully selected depending upon the measurements required.

Table II shows the specification comparison of various instruments from different manufacturers. For example, spectrum analysers of Agilent Technology—models ESA-L series and ESA-E series—are compared. In ESA-L series, three models are shown, which cover different frequency ranges. All models have almost the same sensitivity and resolution bandwidths. Optional lower-resolution bandwidths are also available. In ESA-E series four

models are shown, which cover different frequency ranges. All have the same sensitivity and resolution bandwidth specifications. Optional 1Hz resolution bandwidth is also available. So one has to select a model depending on the frequency of operation and resolution bandwidth requirements.

Another example shown is RF power meter. Power meter from Boonton Electronics Corporation covers different frequency ranges and different power level ranges with different models of power sensors. One has to select a particular sensor depending on frequency range and power measurement range.

Agilent Technologies has a power meter with power sensor covering up to 18 GHz and power range from 1 microwatt to 10 milliwatts. Another meter is a portable handheld model with built-in power sensor covering up to 6 GHz and power range from 1 nanowatt to 100 milliwatts. This also operates on battery. It is suitable for transportable installations where space is a constraint. All power meters have a resolution of 0.001 dB. You can select a power meter that meets frequency and power range requirements of your application.

For example, for a satellite communication earth station operating in C-band frequencies, frequency range up to 6 GHz is sufficient. For stations operating in Ku-band frequencies, frequency range up to 14 GHz is required.

To sum up

While selecting a test instrument, keep in mind present as well as near-future requirements. Accordingly, generate specifications. Compare these specifications with the specifications of instruments available from reputed vendors and narrow down your choices by selecting instruments that meet the required specifications. Now from these options you can select one that costs the least. ●

The author has retired from ISRO as a scientist/engineer